Tenacious-Bench v0.1 — Final Submission Memo

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Page 1 — Executive Summary

Decision

Do not deploy the trained adapter as a standalone generator. Deploy as a rejection-sampling layer in front of the prompt-engineered pipeline.

Headline Result

The trained CPO/SimPO adapter (Qwen 2.5 0.5B) achieved **14.6% pass rate** (6/41 tasks) on the sealed held-out partition.

Metric	Value	95% CI
Delta A: trained vs. baseline	+0.000	[–0.122, +0.122], p = 0.585
Delta B: trained vs. prompt-eng	–0.341	[–0.488, –0.195]
Trained pass rate	14.6%	6/41 tasks
Prompt-eng pass rate	48.8%	20/41 tasks

Delta A is flat (p=0.585). Delta B is negative. Careful prompting of a frontier model outperformed the trained adapter by 34 percentage points.

Cost and Latency Per Task

Component	Cost	Latency
Dataset authoring (API)	$0.003/task ($0.63 total)	—
Preference pair rescue	$0.007 total	—
Training (Colab T4)	$0.00	102s total (3 epochs)
Baseline scoring — DeepSeek V3, n=41	$0.0041	avg 11.45s (med 10.52s)
Prompt-eng scoring — DeepSeek V3, n=41	$0.0041	avg 7.35s (med 6.89s)
Trained adapter inference (Colab)	$0.00	~8–12s (uninstrumented)
Total week spend	**$0.79 / $10 cap**	—

Prompt-eng is 36% faster than baseline (7.35s vs 11.45s) because the 10-rule system prompt produces shorter outputs. Trained latency is uninstrumented; 8–12s estimated from wall-clock (~8 min for 41 tasks on T4).

Production Recommendation

Deploy the prompt-engineered DeepSeek pipeline now — 48.8% pass rate at 7.35s/task, 3.3× the trained adapter's rate at comparable latency. The trained adapter adds value only as a scoring/rejection layer: score every draft, regenerate on fail. For standalone adapter deployment: retrain on 7B backbone with 500+ pairs, Delta A ≥ +0.15 with $p < 0.05$ required.

Page 2 — Skeptic's Appendix

Four Failure Modes Tenacious-Bench v0.1 Cannot Grade

Multi-turn sequence continuation. All 33 F5 tasks test single-touch outreach. No task provides a prior thread as input and requires the agent to reference it without repeating it. An agent that ignores the prospect's reply from email-2 when writing email-3 is invisible to v0.1.

Regulated-vertical constraints. None of the 204 tasks cover healthcare, financial services, or defense companies. HIPAA-adjacent, FINRA, and ITAR outreach constraints are entirely absent. An agent compliant for a Series B SaaS company but non-compliant for a healthcare startup scores identically.

Prospect-reply adversarial override (F3 extension). All F3 tasks embed the adversarial instruction in the initial prompt. None test a prospect reply that attempts to lock in a commitment post-email: "So you're confirming 8 engineers starting the 15th?" — a distinct failure mode from initial-outreach over-commitment.

Competitive displacement framing. No task sets competitor_engaged=true with constraints on what not to say about the competing staffing firm. The schema has a competitor_gap_brief field but no task exercises the forbidden-disparagement constraint flagged by Tenacious delivery leads.

Public-Signal Lossiness in Ground Truth

The rubric's signal_confidence scores are derived from job-posting and Crunchbase data with a systematic 30–90 day lag. A task requiring confident framing at crawl time may require cost-preservation framing (F2) at send time if a layoff occurred in between. This lossiness is directional: layoff events are under-represented in job-posting signals, systematically inflating the F1 pass rate by an unknown amount the rubric cannot correct for.

One Honest Unresolved Training Failure

F1 (Confidence-Unaware Phrasing) regressed 12.5pp after training: 6.2% vs baseline 18.8%. Every F1 pair contained confidence-threshold vocabulary (signal_confidence=0.45, velocity_label=weak) in the prompt. The 0.5B backbone appears to have learned to associate this vocabulary with *assertive* output — the opposite causal direction from the training signal. No hyperparameter adjustment was attempted; the single Colab session was already complete. The F1 regression is unresolved.

Kill-Switch Trigger Conditions

Condition	Measurable Signal	Action
Pass-rate degradation	7-day rolling pass rate on 20-task production sample < 35% (14pp below 48.8% baseline)	Disable adapter rejection layer; revert to unfiltered prompt-eng pipeline
F3 violation	Any confirmed headcount promise for engineers not on bench reaches a prospect	Disable adapter + alert delivery lead
F5 violation	Outreach to opted-out prospect or after 3 touches in 30 days	Disable adapter + alert delivery lead
Systemic failure	7-day rolling pass rate < 20% (below untrained baseline)	Suspend full pipeline; switch to manual review

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All numeric claims trace to ablations/ablation_results.json, training_data/preference_pairs.jsonl, or tenacious_bench_v0.1/dataset_manifest.json.